A Falsifiable Computational Decipherment Hypothesis for the Indus Valley Script: 161 Candidate Proto-Dravidian Anchors and a Three-Slot Positional Grammar

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Abstract

We propose and test a falsifiable computational decipherment hypothesis concerning the Indus Valley Script (IVS, ca. 2600–1900 BCE), which remains undeciphered in the absence of a bilingual text. Working from the Holdat LLC Indus Corpus V3 (1,670 seals, 9 sites, 390 distinct signs, 7,002 sign tokens; Mahadevan M-numbers), we apply positional analysis, bigram/trigram collocational methods, DEDR rebus matching, and syllabic language-model simulation to construct a candidate anchor model for IVS sign readings. We find robust non-random positional structure ($z = 10.3$; no permutation in 2,000 exceeded the observed statistic ($p < 0.001$; 0 of the 2,000 permutations exceeded the observed statistic, so the plain estimate is 0/2,000 and the standard add-one estimate is (0+1)/(2,000+1) = 0.0005, the smallest non-zero value a test of this size can report)), a recurring three-slot positional pattern in inscriptions. The positions are measured; the reading of the three slots as classifier, title, and name/case elements of an administrative vocabulary is this model's interpretation, not a result of the positional tests (§3.2), and propose 161 candidate readings at HIGH or MEDIUM confidence. These readings are assigned to signs that together account for 90.96% of the corpus's sign tokens (69.8% of seals consist entirely of such signs). 'Coverage' here means only that the model assigns a candidate reading to a sign; it does not mean the reading is verified. Four of the 161 readings are provisional (see §2.2); without them the figures are 157 readings and 90.75% of tokens. The corpus itself is not publicly released (see §6), so corpus-level counts cannot yet be re-run by other researchers. A systematic fish-sign isolation test finds 0 isolated fish signs among the 113 fish-bearing seals in the 1,670-seal corpus, across all 9 corpus sites (§3.5). A separate search of the Gulf deposit catalog's text (Laursen 2010) found no fish-sign mentions at all, so it provides no counterexample to that pattern but also no confirmation of it; the two results count different units and are not combined (§3.19). We correct a prior sign-counting error: an earlier count of 268 H+M signs included 107 heuristic placeholders not meeting the stated evidence criteria. Of the 75 HIGH-confidence readings, the reading strings of 44 (59%) also appear in Parpola (1994), the work whose framework guided this analysis — appearances in his text, not a sign-by-sign check that he proposed the same reading for the same sign (see §3.22); the evidence is consistent with a Proto-Dravidian reading framework under the tested assumptions, though Dravidian superiority over proto-Munda has not been formally quantified. The program's internal consistency suite reported 59 of 59 checks passing at Phase-170 (May 2026); on the October 2026 execution tree the changed suite reported 40 passed, 0 failed, 8 warnings (Appendix C). These are internal consistency checks, not external validation. This study does not claim epigraphic finality; it proposes a falsifiable computational model whose method and repository artefacts (scripts, phase reports, and the anchor table) can be re-run and checked. The corpus-level counts in this paper cannot be re-run until the corpus itself is released; that limit is stated here, not only in §6, because most of the paper's numbers are corpus-level counts. The model's structural

predictions and candidate phonetic readings can be tested against larger corpora and future archaeological discoveries. Its corpus-level counts cannot yet be re-run by other researchers, for the reason given above.

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1. Introduction

1.1 The Indus Valley Script Problem

The Indus Valley Civilization (IVC, ca. 2600–1900 BCE) produced several thousand inscribed objects — stamp seals, copper tablets, potsherd graffiti, and the Dholavira signboard — bearing a script of roughly 400 distinct signs in the standard catalogues (this paper's own corpus attests 390 of them; §2.1). Despite a century of scholarship, the script remains undeciphered. The primary obstacles are: (1) the absence of a bilingual text; (2) uncertainty about the underlying language; (3) short average inscription length (~4–5 signs); and (4) limited corpus size relative to modern decipherment corpora.

Competing hypotheses have proposed proto-Dravidian (Parpola 1994; Mahadevan 1977), proto-Munda (Witzel 1999), an Indo-Aryan language ancestral to Sanskrit (Jha & Rajaram 2000 — widely rejected), and a non-linguistic semasiographic system (Farmer, Sproat & Witzel 2004). The Dravidian hypothesis is among the most extensively developed linguistic interpretations: the geographic distribution of Dravidian languages, archaeological continuity between IVC and later South Asian cultures, and a long tradition of applying the "rebus principle" (phonetic punning via sound-alike words) to IVS signs, beginning with Heras in the 1930s and developed most fully by Parpola. Individual rebus readings remain contested; none is treated as established in this paper.

1.2 Prior Computational Work

Rao et al. (2009, *Science* 324:1165) reported that IVS sign entropy falls in the range of known linguistic systems; in their own words, their results "increase the probability that the script represents language" — a probability increase, not a demonstration. That entropy work has been criticised (Sproat 2014) and defended (Rao et al. 2010); the question remains contested, and this paper does not rely on entropy as evidence that the script is a language. Mahadevan (1977) produced the standard sign concordance and M-number catalogue. Parpola (1994, 2010) assembled the most comprehensive Dravidian rebus decipherment. Wells & Fuls (2015) produced an independent sign list and catalogue. This paper's claim is narrower than any priority claim: it reports a candidate model whose assigned readings cover 90.96% of this corpus's tokens, with the sign-by-sign evidence record held in the repository named in §6. No literature search for comparable coverage claims is on record, and none is claimed here.

1.3 This Work

We present a computational grammar and candidate anchor model (Glossa-Lab, Phases 1–170) with the following components and results:

1. A five-layer evidence hierarchy with explicit confidence standards (HIGH/MEDIUM/LOW)
2. 161 candidate readings at MEDIUM or HIGH confidence (90.96% token coverage; 69.8% of seals fully covered by HIGH-or-MEDIUM (H+M) candidates) — Phase-133 corrected prior inflated count; Phase-163 adds 4 provisional sibilant upgrades. 43 of the 161 readings are MEDIUM signs whose recorded evidence is the combined SA+DEDR type; §2.4 gives the bound that removing all 43 would imply (118 signs, 85.28% of tokens), and states why a true SA-free headline cannot be computed from the records

3. A three-slot grammar model empirically derived from positional statistics
4. Two signs assigned DEDR-backed candidate readings for which a Munda derivation was proposed but not established (see §4.2; the recorded basis is DEDR 1709 and DEDR 5388, not substrate evidence)
5. A fish-sign isolation test: 0 isolated among 113 fish-bearing seals across all 9 sites; the Gulf catalog text search found no fish-sign mentions at all — no counterexample, not a confirmation (see §3.19)
6. Correction of M267: genitive particle, not fish sign
7. A publish-ready anchor table covering all 161 H+M candidate readings with DEDR citations (161 candidate readings with DEDR-linked phonetic hypotheses; 107 prior heuristic placeholders removed by Phase-133)
8. Site repertoire divergence measured by KL divergence, with a bootstrap confidence interval for the most divergent pair (Mohenjo-daro–Rakhigarhi; see §3.14)
9. Tamil-Brahmi terminal phoneme comparison (4 of 7 strictly terminal signs match a Tamil-Brahmi terminal category; see §3.18 for the time-gap caveat)
10. Another group's scorecard study (Nair 2026, arXiv:2604.17828), listed here because §3.10 discusses it: it studies a separate digitization of the same published inscription assemblage (ICIT corpus) and does not test this paper's readings or grammar (see §3.10). It is not a component or a result of this work.
11. Parpola overlap count: the reading strings of 44 of the 75 HIGH-confidence readings appear in Parpola (1994) — appearances in his text, not sign-by-sign agreement (see §3.22)
12. Mahadevan text-mining comparison: this program's mining found positional-grammar hits in the Mahadevan papers in its text corpus (per-paper hits in §3.23). These are one scholar's works, counted by this program; this is not independent validation
13. Gulf seal text search: no fish-sign mentions found in the Laursen (2010) Gulf deposit catalog text — no counterexample to the compound-only pattern, and not independent confirmation of it (see §3.19)
14. Sibilant DEDR check (Phase-166): M330=can, M165=cul, M202=can, M198=co — all four PROVISIONAL_MEDIUM, all four DEDR-backed (see §§2.2, 3.27)
15. Expanded Meluhhan name matching (Phase-167): 25 Ur III names tested — no discriminative match; ICIT corpus required for personal names
16. Decode-blocker plausibility check (Phase-168): the existing LOW readings of the 19 top blocking signs are phonotactically plausible. This is a methods check, not a result about the readings; the signs remain LOW, and the check is not counted among the evidence items in §3.30 (see §3.29)
17. Master evidence synthesis (Phase-169): 32 evidence items on this program's own labels — 12 structural, 8 linguistic, 8 external, 4 decipherment — mixing internal consistency checks with external comparisons (see §3.30)
18. Grammar self-consistency retest (Phase-170): 93.2% on a sign-level check that uses different definitions from the Phase-133 measure and is not comparable to it (see §3.31)

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2. Data and Methods

2.1 Corpus

Primary corpus: Holdat LLC Indus Corpus v3 (Miller 2025). 7,002 sign tokens, 1,670 seals, 9 sites: Mohenjo-daro (n=606), Harappa (n=492), Kalibangan (n=110), Dholavira (n=106), Lothal (n=124), Chanhu-daro (n=78), Surkotada (n=61), Banawali (n=60), Rakhigarhi (n=33). Sign encoding: Mahadevan M-numbers. This corpus covers the major excavated IVC sites and represents the primary published formal-seal assemblage.

Sign catalogue: 390 distinct M-number signs attested in the corpus; the anchor table assigns readings to all 397 signs in the Mahadevan catalogue (the additional 7 — M086, M088, M090, M091, M092, M093, M094 — have zero corpus tokens and are included for catalogue completeness). M-numbers are labels, not a count: the 397 catalogue entries carry numbers running from M001 to M416, with 19 numbers unused, so a sign numbered above 397 (for example M402) is normal. The full sign-by-sign inventory, with each sign's corpus frequency, is the file `ANCHOR_STATUS_AUDIT.csv` in the repository. Sign frequencies fall off steeply, from a few very common signs to many rare ones. An earlier version of this paper reported a Zipf exponent of about 1.7 for this distribution and described it as consistent with a natural language writing system. That exponent is this program's own statistic on this corpus; no derivation of it has been published that a reader can check, and we no longer present it as evidence that the script is a language. Yadav et al. (2010), who published a Zipf–Mandelbrot fit for Indus sign frequencies (on a pruned subset, the EBUDS subset, of Mahadevan's 1977 corpus), state the limit themselves: the presence of such a distribution "is not sufficient to declare a sequence as linguistic". A later test of what a fit of this kind can show is reported in §2.1.1 below.

Crosswalk: 25-entry Mahadevan – Parpola crosswalk (mahadevan_parpola_crosswalk.json) for iconographic anchor validation.

Supplementary sources: Crawford (2001), used only for the Saar/Dilmun seal comparanda; Hojlund & Abu-Laban (2012), used only for the Failaka pictorial-fish observation in §3.5; Parpola (1994, 2010), iconographic readings.

2.1.1 What a Frequency-Curve Fit Can and Cannot Show (Spec 068)

We tested what a frequency-curve fit can show, under rules fixed and registered before the test was run (Spec 068, executed as Phase-430; OSF registration 3pbwe). One fitting rule — the Zipf–Mandelbrot rule frozen by Spec 053 — was applied to Indus sign counts and to four comparison systems. The Indus counts came from a different corpus from the one described above: the Shakti corpus, a restricted corpus in the ICIT line (3,045 analysable rows of documented lineage; 11,805 tokens; 557 distinct signs). On those counts the fit gave exponent α = 2.004 and fit error 0.1584 (error is the root-mean-square difference in log space; smaller is a closer fit). On a wider set of rows from the same corpus (documented plus corroborated) it gave α = 1.96 and error 0.1544. The comparison systems were Mesopotamian numerical signs (error 0.5335), heraldic charges from the Dering Roll (0.1839), a synthetic emblem process (median 0.2405), and Linear B (0.9229). Linear B is the one system in the set that is certainly writing, and it fitted the curve worst of the five. Under the rule fixed before the run, that result makes the comparison UNINFORMATIVE: these measures do not separate the one certain writing system from systems that are not writing, so a close Indus fit on them is evidence neither for nor against the script being language. The Shakti numbers are a measurement of the Shakti corpus only. They are not a measurement of the Holdat corpus used in this paper, and they do not confirm or correct any earlier exponent, including the Phase-43 exponent of about 1.35 recorded in this program's files. **The four exponents mentioned in §2.1 and this section must not be compared with each other.** They were measured on different corpora, with different sign inventories and different fitting rules:

Exponent	Corpus	What it is
about 1.7	Holdat (this paper)	This program's own earlier statistic; derivation never published; withdrawn as evidence (§2.1)
about 1.35	an earlier ICIT-lineage layer	Phase-43 record in this program's files; not re-measured here
2.004 (and 1.96 on the wider row set)	Shakti, documented-lineage rows	Spec 068 fit under the frozen Spec 053 rule
2.59	pruned EBUDS subset of Mahadevan 1977	Yadav et al.'s published Zipf–Mandelbrot parameter b

No ranking or agreement between any two of these numbers means anything. Two further limits of the Spec 068 test belong on the record. First, the comparison set contains one certain writing system (Linear B, counted as its 89 syllabogram types in the DAMOS transliterations, 47,292 tokens); the verdict is a verdict about this fitting method on these tables, in which the certain writing system happened to fit worst — it is not a general proof that frequency curves can never carry information. Second, the fit is sensitive to corpus size: on half- and quarter-size samples of the same Shakti rows the exponent fell to about 1.82 and 1.63. The full record of the test — rules fixed in advance, input tables, and outcome — is in this repository at `specs/068-yadav-zipf-controls-successor/` (see `outcome.md` and the `data/` tables), registered before the run at OSF: https://osf.io/3pbwe. Two identification notes belong on the record. First, the comparison systems named in this section are identified in the Spec 068 record (`specs/068-yadav-zipf-controls-successor/` — `outcome.md` and the `data/` tables), which is where a reader will find the test's inputs; this paper gives no §8 entry and no fuller bibliographic identification for the Dering Roll source, the Mesopotamian numerical-sign table, or DAMOS (the source of the Linear B transliterations), and the Shakti corpus and the ICIT line are likewise identified only through that record. Second, a chronology note: 'Phases 1–170' in this paper is the phase sequence behind the May 2026 anchor model. The numbering did not stop at 170 — later work in this program continued the same sequence under spec numbers (Spec 068 was executed as Phase-430, and Spec 071 as Phase-434).

2.2 Evidence Hierarchy

Sign readings are assigned to three primary confidence levels, with PROVISIONAL_MEDIUM used as a flagged subcategory of MEDIUM:

HIGH (75 signs): Two or more of the following evidence criteria are met. The criteria are separate tests, but they are not independent of each other: all except the DEDR and Parpola criteria are computed from the same corpus, and several from the same seals. - Iconographic match: the sign's shape is identified, following the published iconography of Mahadevan and Parpola, as depicting a known object whose Dravidian name yields the reading by the rebus principle (e.g., M045=yānai from elephant icon, DEDR 5175) - Distributional exclusivity: sign appears on one and only one motif type with lift ratio > 5.0 (e.g., M062=erutu exclusive to zebu bull seals) - Terminal marker evidence: sign appears in >95% of cases as inscription-final, matching known Dravidian case suffix (M342=ay/ā, genitive terminal)

MEDIUM (86 signs): Single strong evidence source: - DEDR rebus match with SA-consistency >= 0.15 from syllabic language model simulation (SA: the syllabic language model fitted by simulated annealing, §2.4; for how many signs carry SA in their evidence, and the bound it implies, see §2.4) - Positional profile matching a known morpheme class (INITIAL=title prefix, TERMINAL=case suffix) - Supported allograph of a HIGH/MEDIUM anchor by positional L1 distance < 0.2

PROVISIONAL_MEDIUM (4 signs — M330, M165, M202, M198): Promoted from LOW on a narrower basis than ordinary MEDIUM: at least 3 text-proximity hits in the mined literature (Phase-

163), followed by a within-model check of the DEDR entry, the positional profile, and phonotactics (Phase-166), in which a 'strong' reference count means at least 4 hits — a level M198 (3 hits) does not reach. All four signs remain PROVISIONAL_MEDIUM. The Phase-166 labels CONFIRMED (for M330, M165, M202) and PROVISIONAL (for M198) are that check's internal labels; its own note states that CONFIRMED means only that the reading is phonologically plausible and has literature support, not that it is a HIGH-confidence reading, and no sign's category was changed by them. PROVISIONAL_MEDIUM readings are *included* within the MEDIUM count for coverage accounting but are flagged for expert review before any HIGH-confidence promotion. They are listed separately in Section 3.27. The Phase-163 mining record does not specify the proximity window (how near a reading must appear to a sign's number to count as a hit), the OCR source and version for every mined text, or how ambiguous sign-number matches were resolved. The four promotions therefore rest on a count whose generating rule is only partly recorded; that is a further reason, beyond the absence of expert review, that all four signs remain PROVISIONAL_MEDIUM.

LOW (236 signs): Heuristic assignment only: - Positional assignment (initial/medial/terminal class) - Phase-111 allograph resolution (positional profile match to an existing anchor) - Not used for coverage or "H+M-candidate coverage" computations

NOT COUNTED toward coverage: signs at LOW confidence.

The HIGH, MEDIUM, and LOW labels are rule-based categories defined by the criteria above. They are not calibrated probabilities: no measured error rate attaches to any label. The outcome of the criteria for each sign is recorded in the anchor table (`research/indus/anchor_table.csv`, 'Basis' column). The anchor table records one basis statement and one evidence-type label per sign. It does not record each §2.2 criterion as a separate pass/fail field, so questions of the form 'which signs meet criterion X but not criterion Y' cannot always be answered from the records — §2.4's SA bound is one place where that limit binds, and it is stated there. One limit of the DEDR criterion must be stated plainly. For each sign, the anchor table records only the DEDR candidate that was adopted. How many candidate readings were considered for a sign, which sound changes or reading transformations were allowed when matching, and which candidates were rejected were not recorded; and no multiple-comparison control across the roughly 390 signs and the Dravidian Etymological Dictionary was applied or recorded. A reader therefore cannot compute from our records how often a DEDR match of this quality would arise by chance. The checks that do bear on the wider risk are the corpus tests reported in this paper — the Phase-107 held-out test (§2.4) and the Phase-146 and Phase-150 null results (§§3.11, 3.13) — not a correction to the DEDR search itself. Three of the criteria above are stated at the level this program's records support and no further: the iconographic-match criterion has no written coding protocol, coder-blinding step, or inter-coder reliability figure on record; 'cases' in the terminal-marker criterion means the sign's corpus tokens; and the lift ratio in the exclusivity criterion is computed over the corpus's motif distribution, whose per-motif denominators are in the Phase-132 record's working outputs rather than a published table. A reader cannot re-apply the iconographic or exclusivity criteria to a new sign from this paper alone.

2.3 Positional Analysis

For each sign, we compute: - **Positional profile**: fraction of tokens in INITIAL / MEDIAL / TERMINAL / SOLO positions - **Bigram collocates**: most frequent left and right neighbors - **L1 distance** between positional profiles of candidate allograph pairs

INITIAL-dominant signs are classified as title prefixes or classifier-derived words. TERMINAL-dominant signs are classified as case suffixes or proper-name terminals. MEDIAL signs are classified as content words or personal name syllables.

The headline positional statistic (z = 10.3) comes from the Phase-134 F1 test. We fit the three-slot grammar model to the real corpus (coefficient of determination R^2 = 0.9916) and to 2,000 copies of the corpus in which the signs inside each inscription were shuffled (mean R^2 = 0.4385, standard

deviation 0.05347 — the Phase-134 record's values before rounding). The z-score is the distance between the real value and the shuffled mean, measured in standard deviations: z = (0.9916 − 0.4385) / 0.05347 = 10.344, printed as 10.3. No shuffled copy reached the real value. The same Phase-134 suite also ran a held-out check (F7): positional classes fitted on 1,176 seals predicted the classes of signs on 494 held-out seals with 97.7% agreement, so the grammar model was not only fitted and scored on the same inscriptions. The F7 check behind the 97.7% figure (Phase-134) is: positional classes fitted on 1,176 seals, tested on 494 held-out seals, over the 87 signs attested in both parts; 85 of those 87 signs were assigned the same dominant class in both parts (INITIAL 64/64, MEDIAL 5/5, MIXED 13/15, TERMINAL 3/3), which is the 97.7% printed above. The corpus contains no single-sign inscriptions (Phase-134, test F9: 0 single-sign seals), so the SOLO positional class arises only as a profile category, not from one-sign texts. The permutation check behind z = 10.3 used 2,000 shuffled copies (§2.3); the Phase-134 record gives the count and the null moments but does not record the random seed or the shuffling code's inclusion rules, so the permutation cannot be re-run exactly from the record. The operative definition of this statistic — what the grammar model fits, over which observations the R^2 is computed, and how inscriptions of different lengths enter the fit — is the Phase-134 script (`backend/scripts/phase134_falsification_suite.py`). The Phase-134 JSON record reports the statistic's inputs and outputs (observed R^2, null mean and standard deviation, permutation count) but not the data matrix, and this section does not reproduce the script's procedure. The z-score can therefore be checked arithmetically from the printed inputs (§5), but the fit itself can be re-run only from the script and the corpus.

2.4 Syllabic Language Model (SA Simulation)

For signs without iconographic anchors, we run a simulation (SA: a syllabic language model fitted by simulated annealing) trained on Tamil/Dravidian syllable sequences, and propagate readings through the sign graph. The SA assigns a modal reading (most probable syllable) and a consistency score (fraction of contexts agreeing on the reading). Signs with SA consistency >= 0.15 and a DEDR entry matching the modal reading were promoted to MEDIUM. Two limits must be stated plainly. The 0.15 threshold is a working cut-off chosen by this program; it has never been calibrated against a known script, so its false-positive rate is unknown. And a later test inside this program (Phase-107, October 2026) put the SA machinery itself to a held-out test whose protocol, primary metric, and discrimination rule were fixed in advance in this program's Spec 005 (`specs/005-phase52-v2/spec.md`), recorded in this repository before the runs; no external or public registration of that test was filed. Its agreement on held-out anchors was 0.000, and control runs using a Sanskrit model and a scrambled model could not be told apart from it. On that record, SA agreement by itself carries no information about a sign's value. MEDIUM readings whose only support is SA consistency should therefore be read as unconfirmed candidates; the provenance of every anchor, including which readings have SA in their chain of support, is recorded in the repository (Phase-108 provenance register; Phase-109 re-review). How much of the headline set this affects can be bounded from the audit file. No sign in the 161-set rests on SA alone: every sign whose evidence involves SA also has a DEDR match recorded. 43 MEDIUM signs, accounting for 398 corpus tokens, carry the combined evidence type SA+DEDR in `ANCHOR_STATUS_AUDIT.csv`. If all 43 were removed, 118 signs and 5,971 tokens (85.28%) would remain. That is a bound, not a recomputed headline: the audit file does not record whether any of the 43 also meets a non-SA MEDIUM criterion (a positional profile or allograph support), so a true SA-free headline cannot be computed from the records. The 161 / 90.96% headline therefore stands as the figure for the set as constituted, with this bound beside it. One part of the SA+DEDR criterion cannot be reconstructed from our records: the 'consistency' score is described in the Phase-52 lineage as the fraction of a sign's contexts agreeing on its modal reading, but no record defines how a context is delimited, how contexts are weighted, or how ties are handled. The 0.15 threshold's status is as stated above: a working cut-off, never calibrated. What trained the SA must also be stated at the strength the records support. The records describe its training data only as Tamil/Dravidian syllable sequences, through the Phase-52 lineage of scripts and language-model files. No record we hold names a specific training corpus, a syllable-inventory source, or the size of the training data, so

none is stated here: the SA's training set cannot be reconstructed from this paper or from the phase records.

2.5 Substrate Analysis

For signs resisting Dravidian decipherment (25 signs, Phase-123 — the 25 are listed in the Phase-123 record, `outputs/phase123_munda_substrate.json`, and are not printed in this paper), we check SA modals against: - Munda/Austroasiatic substrate vocabulary (Witzel 1999; Southworth 2005) - BMAC (Bactria-Margiana Archaeological Complex) substrate words (Lubotsky 2001) - Brahui cognates (Dravidian outlier, Balochistan) - Proto-Elamo-Dravidian forms (McAlpin 1981)

Two signs were promoted to MEDIUM in this part of the work (M374 and M351; §§3.3, 4.2). In both cases the recorded basis is a Dravidian Etymological Dictionary entry (DEDR 1709 and DEDR 5388 respectively), reached through the Phase-128 re-review. A Munda derivation was proposed for each, but no Munda source, sound correspondence, or borrowing pathway is established in our records for either, so neither promotion rests on substrate evidence. No general standard for what counts as a substrate match was fixed in advance; substrate proposals are therefore reported in this paper as proposals, not as a promotion route. Of the 25 signs checked, 2 resulted in candidate readings (M374 and M351; §§3.3, 4.2).

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3. Results

3.1 Sign Anchor Summary

Confidence	Signs	Token coverage (%)
HIGH	75	70.34% (4,925 tokens)
MEDIUM (excluding provisional)	82	20.41% (1,429 tokens)
PROVISIONAL_MEDIUM (counted within MEDIUM)	4	0.21% (15 tokens)
MEDIUM (including provisional)	86	20.62% (1,444 tokens)
LOW	236	9.04% (633 tokens)
H+M total	**161**	**90.96%**

Note: The confidence table covers all 397 Mahadevan catalogue signs (75 HIGH + 86 MEDIUM + 236 LOW); the corpus attests 390 of these. The 7 catalogue-only signs (zero corpus tokens) are included for completeness and do not affect token or seal coverage metrics. The MEDIUM count includes 4 PROVISIONAL_MEDIUM signs (M330=can, M165=cul, M202=can, M198=co), included for coverage accounting but flagged for expert review before any HIGH-confidence promotion.

The 161 H+M anchors (75 HIGH + 86 MEDIUM) cover 6,369 of 7,002 corpus tokens (90.96%). The remaining 633 tokens (9.04%) have LOW-confidence readings only. Token counts by sign and category are recorded in `ANCHOR_STATUS_AUDIT.csv` in the repository, and the full anchor list is `research/indus/anchor_table.csv`. The four PROVISIONAL_MEDIUM signs (§2.2) account for 15 of the 6,369 tokens; without them, 157 anchors cover 6,354 tokens (90.75%), a figure independently recounted in Phase-132. Counts are from `ANCHOR_STATUS_AUDIT.csv` and sum to the corpus total of 7,002 tokens. Note: a prior count of 268 included 107 heuristic placeholder assignments removed by Phase-133.

3.2 Grammar Model

Statistical analysis of inscription structure identifies a three-slot pattern that is consistent at the level of individual signs' positions. Whole inscriptions match the full pattern less often: in the Phase-133 check, 5.1% of seals matched fully and 38.4% partially (§3.31). 'Consistent' in this paper refers to the sign-level regularity unless a sentence says otherwise:

```
[SLOT 1: INITIAL] - [SLOT 2: MEDIAL] - [SLOT 3: TERMINAL]
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Positional statistics establish the three positions and which signs favour each. The meanings attached to the positions below are this model's interpretation, not outputs of the statistics: we interpret Slot 1 as an animal classifier (from the seal motif), Slot 2 as a guild or group title, and Slot 3 as a personal-name ending and case marker. The pattern was tested against inscriptions with their signs shuffled; it was not tested against rival templates such as owner marks, commodity marks, or clan emblems, which could also produce fixed position patterns.

Slot 1 (INITIAL): Examples: - Unicorn seals: M211=kol (lord/unicorn, from kōl+rebus) - Zebu bull seals: M062=erutu (bull/ox), M073=kōṉ (king/bull) - Elephant seals: M045=yānai, M039=āṉai - Rhinoceros seals: M060=kāṇṭāmirukam (a descriptive compound gloss, 'rhinoceros'; no DEDR entry is recorded for this sign — its support is iconographic and distributional only, 20 tokens)

Slot 2 (MEDIAL): Examples: - M099=kol/koḷ (vessel/jar, guild-of-vessels) - M233=ūr (settlement, of-the-town) - M162=il/iḷ (house/dwelling) - M261=muruku (young man; 2 tokens only, from Parpola 1994 via this program's Phase-51 — the further gloss 'deity Murugan / guild-of-Murugan' is not supported by anything in our records and is withdrawn)

Slot 3 (TERMINAL): Examples: - M342=ay/ā (genitive terminal, most frequent terminal at 584 tokens) - M176=an/aṇ (masculine suffix) - M367=am (neuter suffix) - M336=iṉ (locative case marker)

Under this interpretation, an inscription would encode 'the [animal-totem guild title] [person's name]'. The model predicts the *position* of the name element only; it does not read any person's name (§3.9, §4.3). The pattern is comparable in function to a medieval European guild seal identifying a craftsman by occupation and personal name — an analogy, not evidence.

3.3 Selected High-Confidence Readings

HIGH confidence (selected readings): HIGH is assigned when any two of the §2.2 criteria are met; it is not an iconography-only category. In this table, most rows rest substantially on iconography or motif exclusivity, but M342 and M176 rest substantially on their terminal distributions, and M261 on a published source (Parpola 1994, via Phase-51). The audit file records two evidence types for each of these three signs — iconographic and distributional (`ANCHOR_STATUS_AUDIT.csv`, Evidence-Type ICONOGRAPHIC+DISTRIBUTIONAL) — and that recorded pair, not any single row of this table, is the basis of their HIGH status under the two-criterion rule in §2.2. Two cautions belong with that statement. The audit's two types are recorded as types, not scored criterion by criterion, and they do not map one-to-one onto the three criteria named in §2.2. And M176's printed terminal figure — 94% of its 356 tokens inscription-final — does not by itself meet §2.2's terminal-marker criterion (>95%); its HIGH status rests on the audit's recorded evidence pair, stated here so that a reader applying §2.2 to this table alone is not misled.

Sign	Reading	DEDR	Basis
M045	yānai	5175	Elephant icon, Tamil yānai
M062	erutu	820	Exclusive to zebu bull (lift > 5.0)
M073	kōṉ	2206	Exclusive to zebu bull (lift > 5.0)

Sign	Reading	DEDR	Basis
M342	ay/ā	—	Candidate case suffix: 584 tokens, 96% of them inscription-final; the case-suffix label is an interpretation of that distribution, not a demonstrated paradigm
M176	an/aṇ	135	Candidate masculine suffix: 356 tokens, 94% of them inscription-final; the gender label is likewise an interpretation of distribution. HIGH on the audit's recorded iconographic + distributional evidence pair (see the note above); the 94% figure alone does not meet the >95% terminal-marker criterion
M211	kol	2173	Unicorn seals exclusively
M125	vil	5407	Bow iconography, Tamil vil (bow)
M261	muruku	4988	Wheel/spindle (2 tokens; the reading is *muruku*, 'young man' — the further gloss 'Tamil Murugan' is withdrawn, as in §3.2)
M264	peN	4394	Female figure, Tamil peN (woman)
M328	āl	340	Man figure, Tamil āl (man/person)
M391	ka/kaṇ	1166	Numeral stroke, Tamil kaṇ
M059	ēḷ	912	Seven strokes, Tamil ēḷ (seven)

MEDIUM confidence (substrate):

Sign	Reading	DEDR	Substrate	Basis
M374	kul	1709	DEDR 1709 (kulam, clan/lineage).	A Munda *kul* has also been proposed by this program (Phase-128); no Munda source for it is cited in our records, and the DEDR entry alone does not make the reading a Munda loan.
M351	vī	5388	DEDR 5388 (vī, seed).	A derivation from Munda *bi* (seed/sprout) by a 'bilabial shift' has been asserted in this program's notes; no regular sound correspondence or borrowing pathway has been shown, so the Munda derivation is unestablished.

MEDIUM confidence (positional):

Sign	Reading	DEDR	Position	Basis
M072	mā	4751	INITIAL	All 12 tokens INITIAL; mā=great; collocates peN/veL
M149	or	987	MEDIAL	MEDIAL between kōṉ and -an; oru=one/a certain
M185	pul	4336	MEDIAL	MEDIAL descriptor; pul (humble/grass)

3.4 Major Correction: M267 Is Not the Fish Sign

Version 6 of the anchor table assigned M267 (corpus frequency=400) as "mīn/fish." This was incorrect. M267 appears across all nine motif types with no iconographic specificity. The counts are the Phase-132 record's (test V03 — the distribution behind the chi-square printed below): unicorn 147 tokens, zebu bull 78, elephant 43, rhinoceros 26, buffalo 20, script only 34, geometric 19, tiger 17, gharial 16 — 400 tokens in total, the sign's full corpus frequency. (Counts printed in earlier drafts — unicorn 127, zebu 72, elephant 37, rhinoceros 25 — came from the anchor file's basis note, do not match the test record's table, and are withdrawn; the chi-square below was never computed from them.) It is distributed at 5.7% of all tokens. The actual fish sign is M047 (frequency=13, P47 in Parpola's system), verified by crosswalk. M267 is a high-frequency functional/suffixal sign. Of its 400 occurrences, 324 (81.0%) are MEDIAL, 59 TERMINAL, and 17 INITIAL; its most frequent right-hand neighbour is M099 (84 occurrences), which is the basis for describing its distribution as pre-title. Its distribution across seal motifs does not differ significantly from the counts expected if its 400 tokens were spread across the nine motif types in proportion to each type's overall share (chi-square = 12.98, p = 0.1124, 9 motif categories, 8 degrees of freedom; the full nine-motif count table is in the Phase-132 record, `backend/reports/phase132_validation_report.json`, test V03), which is why a motif-bound reading such as 'fish' fails and a grammatical-particle reading is preferred. ('Uniform' in the Phase-149 summary of this test is that record's shorthand for the same expected-count comparison.) Its current MEDIUM reading is iN/in (genitive particle). This correction is methodologically important: it demonstrates the risk of reading frequency alone as a phonetic indicator. The same warning applies to our own grammatical labels for M342 and M176 (§3.3), which also rest substantially on position and frequency and are stated there as candidates, not identifications.

3.5 Fish Sign Isolation Test

The fish-sign model this test was built on is Avishai Roif's, published in Roif (2025a, 2025b). In his model the fish sign is read /mi/ (mīn, "fish/star") as a value and divine marker and as a guild/title marker (Roif 2025a); in the expanded treatment the fish sign's holder is a "coastal controller" — a maritime guild leader — and the fish, boat, and jar signs recur predominantly at port sites, including Lothal (Roif 2025b). The predictions this program tested — a fish sign that functions as an initial/title marker and is never isolated, and fish signs as maritime markers rather than solitary quantity markers — are Roif's published predictions. The measurements reported in this section and in §3.19 (the counts, site tables, and verdicts) are this program's own, from our corpus work: we do not attribute our statistics to Roif, and his model is not ours.

We tested whether fish signs appear in isolation anywhere in the formal stamp-seal corpus. Testing the full fish sign family — M047, M049, M052, M053, M054, M055, M056, M145, the signs this program's Mahadevan–Parpola crosswalk file lists as fish-shaped — across all 9 sites. This printed list is the family's citable form: the crosswalk file (`research/indus/mahadevan_parpola_crosswalk.json`) does not mark the family as a separate list.

Site	Type	Fish seals	Isolated	Compound
Lothal	coastal port	6	0 (0%)	6 (100%)
Harappa	inland	33	0 (0%)	33 (100%)
Mohenjo-daro	inland	35	0 (0%)	35 (100%)
Dholavira	inland	11	0 (0%)	11 (100%)
Kalibangan	inland	13	0 (0%)	13 (100%)
Chanhu-daro	inland	5	0 (0%)	5 (100%)
Banawali	inland	4	0 (0%)	4 (100%)
Rakhigarhi	inland	3	0 (0%)	3 (100%)
Surkotada	inland	3	0 (0%)	3 (100%)

Site	Type	Fish seals	Isolated	Compound
Total		**113**	**0 (0%)**	**113 (100%)**

A fish sign counts as 'isolated' if it is the only sign in its inscription; the count is of fish-bearing seals (113), and the occurrence-by-occurrence listing is in the supplemental file `fish_sign_compound_context.csv`. The fish sign is **never** isolated in the formal stamp seal corpus, at any site, including Lothal (a major coastal site, where a large basin has traditionally been interpreted as a dock — an interpretation some archaeologists dispute). The compound-only pattern holds at coastal and inland sites alike. That rules out a purely local, single-site habit; it does not by itself prove a scribal rule, because this corpus contains formal stamp seals only, and selection or object-type effects could produce the same pattern. This result is compatible with a range of interpretations of the fish sign's function; the present study does not adjudicate between them. What it establishes is a fact about this corpus — 0 isolated fish signs among its 113 fish-bearing formal stamp seals — and, because the corpus is one private collection of formal stamp seals only, not yet a fact about the Indus script at large. The fish-family list printed above is the family's citable form; no record gives a published page or figure for each sign's fish-shaped classification in the crosswalk, and none is claimed here.

A word on what 0 can and cannot show at these sizes. With 0 events in 113 seals, the pooled isolated rate is bounded at about 2.6% (one-sided 95% bound for zero events). At the smallest sites the bound is far weaker: at Lothal (6 fish-bearing seals) 0 events bounds that site's rate only below about 39%, and at Rakhigarhi and Surkotada (3 each) below about 63%. 'Site-invariant' in this paper therefore means only that no site showed an exception — not that every site's rate is tightly measured.

A separate, weaker observation concerns pictures rather than script: Dilmun-style seals at Failaka Island (Hojlund & Abu-Laban 2012) show fish in compound *pictorial* contexts (human holding fish, fish attached to staffs), never as a solitary motif. This program's records do not tie the observation to specific catalogue numbers, figures, or pages in Hojlund & Abu-Laban (2012); it is reported as a general observation from that volume, not as a checked count. Pictorial motifs are not Indus graphemes, and this is motif-level context only, not script evidence.

3.6 Seal Coverage Statistics

Metric	Value
Total seals	1,670
Fully covered by H+M candidate readings	1,165 (69.8%)
Blocked by LOW-confidence signs	505 (30.2%)
Blocked by unknown signs	0 (0%)

No site-by-site coverage table for the 161-sign set exists: it was never computed, in this paper or in the program's records. For the earlier 157-sign set, the Phase-133 record does give fully covered seals by site, from 55% (Kalibangan) to 79% (Surkotada); Rakhigarhi (n=33) was 64% and Dholavira 72% on that set. Nothing about the 161-set's distribution across sites should be inferred from the 157-set table.

The remaining 505 seals not fully covered by H+M candidate readings contain at least one LOW-only sign (§3.6 table). Of these, 18 signs with corpus frequency 5–6 resist MEDIUM promotion entirely (Appendix B); together they account for 92 tokens. The Phase-144 record analyses them as a MEDIAL-position class; no record shows their distribution across sites or motif types, and at frequencies of 5–6 no distribution across 9 sites and 9 motif types could be 'equal' in any case. Within the model they are personal-name syllable candidates; the statistics do not distinguish them from any other medial-only sign class (§3.9), and their assignment would require a larger corpus or a bilingual text.

3.7 Collocate Network and Formula Structure (Phase-142)

From the full corpus (1,670 seals, 7,002 tokens), we extracted 2,647 distinct bigrams — adjacent sign pairs within inscriptions, in reading order — including 1,485 H+M-by-H+M bigrams; the full table is the supplemental file `formula_bigram_table.csv`. The dominant formula backbone is **M342·M176** (*ay/ā·an/aṇ*), which appears on 122 seals (7.3% of corpus, PMI=2.43) — the highest-frequency H+M-by-H+M bigram pair. (It is not the highest-PMI pair in the corpus; the highest PMI values belong to rare pairs with counts of 3–5.) The next cluster is M267·M099 (*iN/in·kol/kol*, 84 seals), M099·M342 (81 seals), and M176·M099 (74 seals). 97 distinct formula types are identified, organised by INITIAL sign.

Two terms used in this section are defined here. PMI is pointwise mutual information, in bits: $PMI(a,b) = \log_2[P(a,b)/(P(a) \cdot P(b))]$, where P(a, b) is the pair's share of all bigram tokens and P(a) and P(b) are the two signs' shares of all sign tokens (Phase-142 script, `backend/scripts/phase142_collocate_network.py`). For the worked example in this section: M342·M176 occurs 122 times; M342 accounts for 584 of the corpus's 7,002 sign tokens and M176 for 356 (§3.3); the pair's share is taken over all bigram tokens in the Phase-142 record (`backend/reports/phase142_collocate_network.json`), which is also where the distinct-bigram counts above (2,647 in total; 1,485 H+M-by-H+M) are recorded. PMI values from this section use sign-token marginals, as defined above, and are not comparable to bigram-marginal PMI values in other work. A 'formula type' is one INITIAL sign together with the seals whose inscriptions begin with it: the 97 types are the 97 distinct INITIAL signs found among the H+M-covered seals (Phase-142 formula classification) — which is why §3.12's clusters are organised by INITIAL sign.

Positional constraint test (Phase-142D) — descriptive result only: in the raw counts, 17 of the 21 tested H+M signs have different collocate sets when they stand in INITIAL position from when the same signs stand in non-INITIAL positions. Whether that difference is more than chance is what the permutation test in §3.13 was for; its verdict was INCONCLUSIVE. The 17/21 figure is therefore a description of the counts, not evidence for or against position-specific collocates, and we do not use it as support for the grammar model. M267 is the strongest single case on the descriptive numbers: 324 of its 400 occurrences (81.0%) are MEDIAL (§3.4).

3.8 Iconographic Cross-Encoding (Phase-143)

We tested 394 INITIAL-sign $\times$ iconography pairs for enrichment (chi-square, one degree of freedom per pair, Bonferroni-corrected threshold $\chi^2 \approx 14.7$). The supplemental file `iconographic_formula_pairs.csv` contains the 15 strongest enriched pairs, in the Phase-143 record's order; it is not the full 394-pair table, and the full table — including the 63 passing pairs and the pairs that did not pass — exists only in the Phase-143 record's working outputs, not as a published file. The 394 pairs are the pairs formed by the INITIAL signs and icon types attested together in the corpus under the Phase-143 script's inclusion rule; the exact inclusion rule is not stated in the record, so the Bonferroni denominator of 394 is reported as recorded and cannot be re-derived from the published files. **63 pairs (16%) pass the corrected threshold**; an adversarial recheck (Phase-149) estimates about 43 of the 63 are robust, the rest marginal, and several expected counts are small (as low as 1.0), so the largest associations are the reliable ones. The chi-square printed for each pair is that pair's single-cell contribution, $\chi^2 = (O-E)^2/E$, computed from the unrounded expected count E — the count expected if sign and icon were independent, from the pair's row and column totals (Phase-143 script). It is not a full 2×2 chi-square for the pair. The unrounded expected counts are on record for the first three rows (2.036, 2.874, 2.754); for M067 and M039 the record gives the expected counts to two decimals only (1.53, 2.16), so those two χ^2 values do not recompute exactly from the printed table. 'Robust' in the Phase-149 recheck means the pair stands at the Bonferroni threshold for 394 tests ($\chi^2 \gtrsim 14.7$); the remaining roughly 20 of the 63 are about the number of false positives and marginal pairs expected when this many pairs are tested (Phase-149, challenge C2). Strongest associations:

INITIAL sign	Reading	Icon	Observed	Expected	chi-square
M060	kāṇṭāmirukam	rhinoceros	20	2.04	158.5
M045	yānai	elephant	24	2.87	155.3
M016	kaḷiṟu	elephant	23	2.75	148.8
M067	kōṭṭāṉ	rhinoceros	15	1.53	118.9
M039	āṉai	elephant	18	2.16	116.5

These are the five largest associations in the Phase-143 record and in the supplemental file, in order. The next associations are M068 (maṟi, rhinoceros, χ^2 = 103.0), M062 (erutu, zebu, 84.6) and M073 (kōṉ, zebu, 75.5); M059 (ēḷ/eḷ, unicorn, 66.9) ranks ninth. An earlier draft printed M062 and M059 as the fourth and fifth rows; no selection rule for that choice is on record, and the table is corrected to the record's order here.

For the enriched subset of pairs — above all the largest associations — this shows that the inscription's INITIAL sign and the carved animal on the same seal are co-selected, not independent decorations. It shows nothing about the pairs that did not pass the threshold. It does not by itself show what either channel means: in particular, for signs whose reading was itself suggested by the animal motif, 'the reading matches the animal' is circular and is not claimed here as validation. Whether the co-selection encodes professional identity is an interpretation (§4.1), not a result of this test.

3.9 Blocking Sign Analysis and Structural Ceiling (Phase-144)

All **top blocking signs** (those preventing H+M-complete coverage of the most seals) are 100% MEDIAL (m_rate=1.0, i_rate=0.0, t_rate=0.0). Within this model, a sign class that appears only in MEDIAL position is what personal-name syllables would look like; it is also what any medial-only sign class would look like, and the statistics alone do not distinguish the two. In an identity-encoding system: occupational titles (INITIAL) and case markers (TERMINAL) are recoverable from distributional statistics and iconographic anchors, but personal name components (MEDIAL) are arbitrary and unrecoverable without a bilingual text. The 30.2% of seals not fully covered has not been resolved by the methods used here. That is a limit of the current method and corpus, not a proof that the ceiling is structural: the same blocking signs carry rich collocate context (Phase-144), and a larger corpus or a bilingual text could move it.

3.10 Cross-Corpus Structural Comparisons (Phase-145 + Nair 2026)

Comparison with the CISI corpus (179 inscriptions, Parpola P-numbers, mayig digitization): the Phase-145 record reports 87.08% for the share of CISI inscriptions opening with an INITIAL-class sign. The number of multi-sign inscriptions behind that percentage, and the count opening with an INITIAL-class sign, are recorded nowhere — not in the Phase-145 report and not in any other record we hold — so the percentage cannot be checked or recomputed, and we report it as an uncountable figure carried from that record, not as a verified result. CISI is the Corpus of Indus Seals and Inscriptions; neither it nor the mayig digitization has a §8 entry, and this paper can give no fuller bibliographic identification for either from its records. The positional-class distributions of the two corpora differ substantially (KL divergence = 0.591; the Phase-145 record itself rates structural agreement LOW — for example, MEDIAL-class share is 68.75% in CISI against 34.8% in Holdat). Because the two corpora use different sign numberings and different coverage, we report the difference without explaining it away. The opening-sign figure (87.08%, above) is the only quantity in this comparison that points the same way in both corpora, but because its numerator and denominator are not on record it is reported here as an uncountable carried figure, not as a point of agreement.

Independent scorecard study (Nair 2026, arXiv:2604.17828): Applying a multi-metric discrimination framework to 1,916 deduplicated ICIT inscriptions (584 unique signs, 11,110 tokens), Nair reports a

Zipf slope of -1.49 (a signed regression slope; the exponents in §2.1.1 use the opposite sign convention and a different fitting rule, and the two are not comparable), conditional entropy of 3.23 bits, and a permutation result of 0 exceedances in 1,000 trials (reported by Nair as percentile 0.000). He reports matching his administrative baseline on two of his four metrics; the fourth metric is not one this paper uses. Nair's corpus is a separate digitization (G-prefix ICIT coding) of the same published inscription assemblage, analysed with a different methodology; it is independent work, but it is not an independent body of inscriptions; his permutation test concerns within-inscription ordering, a different statistic from our Phase-134 F1 positional test ($z = 10.3$; no permutation in 2,000 exceeded the observed statistic ($p < 0.001$; 0 of the 2,000 permutations exceeded the observed statistic, so the plain estimate is 0/2,000 and the standard add-one estimate is (0+1)/(2,000+1) = 0.0005, the smallest non-zero value a test of this size can report)). Nair's own conclusion is deliberately hedged: the Indus corpus occupies an intermediate position between his synthetic non-linguistic baselines, matches the administrative baseline on only two of four metrics, and, in his words, the study does not settle the linguistic question. One of Nair's four measures, his Zipf slope, is a frequency-curve fit of the kind tested in §2.1.1: on the frozen comparison reported there, that kind of fit did not separate writing from non-writing, so we do not use his slope — any more than our own earlier figure — as evidence about linguistic status. His results are consistent with non-random sequential structure in the Indus sign system; they do not test the positional grammar, the anchor readings, or the Dravidian hypothesis advanced here.

3.11 Phonological Exclusivity Test (Phase-146)

F3 redesign using phonemes that occur in Dravidian but are absent from the Sanskrit inventory used in this test (ḷ U+1E37 retroflex lateral; r̲ U+1E5F alveolar trill; n̲ U+1E49 alveolar nasal; l̲ U+1E3B retroflex approximant). Results on the 157-sign H+M set: **0/157 readings contain the test's Sanskrit-side markers** (aspirated stops, palatal nasal ñ, visarga ḥ); **35/157 (22%) contain Dravidian-side markers**. The test's recorded verdict is **INCONCLUSIVE**, and we report it as such: the share of readings carrying Dravidian-side markers (22%) is below the rate the test's own null model expects (35%; one-tailed binomial $p = 0.9996$ — the upper tail, the probability of a count of 35 or more in 157 readings at the null rate 0.35, computed in the Phase-146 script by a normal approximation; the observed count of 35 sits well below the null expectation of about 55, which is why the value is close to 1), and the readings tested were themselves chosen from Dravidian sources (DEDR), so an absence of Sanskrit-only sounds in them is partly built into how they were selected. This test does not establish Dravidian exclusivity. Both sound inventories used in this test are the test's own working lists, printed in full in this section and existing nowhere else in this program's records; no published Sanskrit or Dravidian phoneme-inventory source is cited, because none was used. The test's result (INCONCLUSIVE, §3.11) is therefore a result about these lists as much as about the readings.

Note: this test was run on the 157-sign set (pre-Phase-163). The four later sibilant additions all begin with c-, which is a valid Proto-Dravidian onset (Krishnamurti 2003) but is not Dravidian-exclusive — Sanskrit also has c-initial words. Counting them, the figures would be 39/161 (24.2%) readings with Dravidian-side markers and 0/161 with Sanskrit-side markers, under the same limits just stated. The reading set counted here is the same set whose V07 phonotactic violation rate is 23.4% (§3.18); nothing in this section's result offsets that rate.

3.12 Formula Semantic Clustering (Phase-148)

Grouping the 97 INITIAL-sign clusters — the 97 distinct INITIAL signs found among the H+M-covered seals, as defined in §3.7, with no further size threshold applied to the grouping, by the semantic domain this model assigns to each cluster's proposed reading:

Domain	Clusters	Seals	% corpus
UNRESOLVED	51	845	50.6%
ANIMAL_GUILD	25	446	26.7%
MATERIAL	6	145	8.7%
DEITY_TITLE	9	135	8.1%
CIVIC_ROLE	6	99	5.9%

The domains are assigned from the proposed readings themselves, so the domain shares describe how this model's own readings are distributed; they cannot be evidence for the readings. In particular, the ANIMAL_GUILD share (26.7%) follows from the grouping rule and is not an independent finding about guilds.

All domains share the same terminal sign profile (M342/M176 dominant) — a measured fact about the model's own sign assignments, consistent with a common positional skeleton across the model's domains. Whether the domains were registers is not shown by this or any other test in this paper (see the note below). The ANIMAL_GUILD domain (animal-named INITIAL signs paired with matching animal icons) accounts for 26.7% of seals — consistent with a totem-based guild identity system. CIVIC_ROLE (ūr/settlement, pār/look, koḷ/vessel) and DEITY_TITLE (vēl, nal, nēr) clusters are interpreted by this model as administrative and devotional groupings. No test in this paper shows that they functioned as registers. The per-cluster list (97 clusters, with each cluster's seals and terminal profile) is in the Phase-148 record (`backend/reports/phase148_formula_semantics.json`).

3.13 Polysemy Permutation Null (Phase-150)

A permutation null test (n=1,000 within-seal position shuffles) was run against the Phase-142D polysemy finding. Observed collocate divergence rate: 66.7% of the 21 tested signs (listed in the supplemental file `polysemy_divergence_summary.csv`; the signs are those attested in more than one positional slot) show KL > 0.3 bits between INITIAL and non-INITIAL collocate profiles. Null distribution mean: 80.7% (SD=0.088). The observed rate is *below* the null mean (z = -1.6; one-tailed permutation p = 0.955). The test's recorded verdict is **INCONCLUSIVE**: on this measure, the real corpus does not show more position-specific collocate divergence than shuffled corpora, so the Phase-142D descriptive result (81% of the tested signs show positionally distinct collocate sets) cannot be claimed as above-chance evidence. We report the test because it was run; it neither supports nor refutes the grammar model. The support for positional grammar in this paper rests on the Phase-134 tests (§2.3, §3.10), not on this one.

3.14 Site Repertoire Divergence Bootstrap (Phase-151)

In this section, KL divergence compares two sites' sign repertoires: each site is represented by the distribution of its H+M sign tokens over the signs attested at that site. The measure is symmetric KL — half the sum of the two one-way divergences, $D_{KL}(P|Q)$ and $D_{KL}(Q|P)$ — in bits (log base 2), with Laplace smoothing (one count added for every sign in the pair's combined vocabulary), as defined in the Phase-151 script (`backend/scripts/phase151_site_kl_bootstrap.py`). The bootstrap resamples seals with replacement within each site, 1,000 resamples (the Phase-151 script fixes the random seed at 42); a confidence interval was computed for the most divergent pair only. Neither the Phase-151 record nor its script names the interval method (percentile, basic, or bias-corrected), so the printed interval is reported as recorded, without a method name. The KL divergence in §3.10 is a different quantity: a one-directional divergence, $D_{KL}(\text{Holdat}|\text{CISI})$, over the four positional-class shares, with 0.001 added to each share (Phase-144/145 script). The 36 observed site-pair values in this section are descriptive; intervals are printed for one pair only, so no family of interval claims is made.

Bootstrap confidence intervals (n=1,000 resamples) were computed for the most divergent pair only. Key findings: Rakhigarhi (n=33) has the largest observed divergence against every other site in this corpus. All 36 observed site-pair values are in the Phase-151 record (`backend/reports/phase151_site_kl_bootstrap.json`); the largest is Mohenjo-daro–Rakhigarhi (0.509), and the next largest is also a Rakhigarhi pair (Harappa–Rakhigarhi, 0.481), so the 'most divergent pair' named below can be checked against that record. Intervals were computed for that one pair only, so the ordering of the other pairs is descriptive, not interval-tested. A bootstrap confidence interval was computed for the most divergent pair: Mohenjo-daro vs Rakhigarhi KL=0.509, 95% CI [0.483, 0.750]. The other Rakhigarhi pairs have observed KL values between 0.250 and 0.481 (Harappa 0.481, Kalibangan 0.350, Lothal 0.348, Dholavira 0.320, Chanhu-daro 0.291, Surkotada 0.290, Banawali 0.250); no confidence intervals for those pairs are on record, and the earlier draft's claim that five Rakhigarhi comparisons all have interval lower bounds above 0.3 is withdrawn. The large-site pair (Mohenjo-daro vs Harappa) shows KL=0.070 — a small measured difference between the two repertoires. A shared scribal tradition is one possible explanation; corpus composition and object-type selection were not controlled for, and could produce the same small difference. Rakhigarhi has only 33 seals; resampling those 33 seals cannot add information about the site, so its distinctiveness should be read as a measured difference on a small sample, not as proof that the difference is not a small-sample effect.

3.15 Shu-ilishu Phonological Test (Phase-152)

The Shu-ilishu seal (Ur III, c.2020 BCE) is the one external name this program has tested against: its owner is named "Shu-ilishu" (ŠU-i-li-šu) in the seal's Akkadian cuneiform inscription, and his title is "interpreter of the Meluhha language." The name is Akkadian, the seal carries no Indus inscription that we read, and the test is therefore narrow: it asks only whether our candidate reading inventory contains signs for the sounds of his name. Testing the four phonological slots /su/, /i/, /li/, /shu/ against the pre-Phase-163 157 H+M readings: 2/4 slots are covered (/i/ by the candidate signs the Phase-152 record lists — M267, M162, M336, M305, M045 — of which M305's recorded reading, ōṭu/comitative, contains no /i/ sound; the record's slot list is reported here as recorded, and this mismatch is one reason the test's result is stated as inventory overlap only; /li/ by 4 signs including M162=il/iḷ). No seal in the corpus matched three or more of the four slots (Phase-152: candidate seals = 0). The sibilant /su/ phoneme is absent from the current H+M reading set. Result: NO_SEAL_MATCH — partial inventory overlap only (2 of 4 slots covered; candidate seals = 0). The Phase-152 record labels this outcome PARTIALLY_SUPPORTED; we do not use that label, because every seal-level result of the test is zero and a label containing 'supported' invites misreading even with a qualifier. The critical gap is coverage of /su/ and /shu/. The external facts in this section (the Ur III date, c. 2020 BCE; the title 'interpreter of the Meluhha language'; the provenance recorded as 'excavated at Ur, held at British Museum') are as recorded in the Phase-152 record (`backend/reports/phase152_shu_ilishu.json`), which is the source this section relies on; no §8 entry covers them. This test was run against the 157-sign H+M set before the Phase-163 sibilant additions. Phase-166 partially addressed the broader sibilant gap by adding M198=*co* as PROVISIONAL_MEDIUM, but it does not cover /su/ or /shu/. A larger corpus remains necessary for this external anchor battery. The inventory checked in this section is the same reading set whose V07 phonotactic violation rate is 23.4% (§3.18).

3.16 Sibilant Anchor Inventory (Phase-153)

A targeted audit of sibilant-initial Dravidian readings in the current H+M set finds 6 signs whose readings begin with a ca-/ce-/ci-/co- sound (M025=cem, M078=cēr, M066=cōḷ, M028=cōḷ, M237=ce, M118=car). An earlier count of 7 included M305, whose reading is ōṭu (the comitative suffix; §3.18) — it has no sibilant-initial sound and is removed from this count. The Phase-153 record asserts a count of 7 without listing the signs; on the readings printed in this paper and recorded in the audit file, 6 is the checkable figure. The /su/ or /shu/ phoneme is not covered. 17 DEDR-attested Dravidian

sibilant roots were identified as candidate readings for unanchored MEDIAL signs. The /su/ gap is targeted for the ICIT corpus expansion (5,318 inscriptions vs 1,670) — a larger corpus provides more distributional evidence for low-frequency sign assignments. The 17 roots, with their DEDR numbers and glosses, are listed in the Phase-153 record (`backend/reports/phase153_sibilant_anchors.json`).

3.17 Vowel Harmony Methodology Resolution (Phase-154)

The V12 vowel-harmony warning is unresolved, and our own records disagree about it. Depending on the version of the test, the pass rate is 75.3% (Phase-132: 476 of 632 decoded seals, against an 85% threshold), 64.2% on a stricter cross-reading version (Phase-133e), or 36.1% (Phase-154: 417 of the 1,154 seals checked *passed*; 737 seals failed), where the recorded verdict is V12_ACTIVE_CONCERN. The 418 figure in that record is a count of word-internal (root) violations, not a count of seals — a different unit, close to the 417 passing seals only by coincidence: 418 root violations plus 319 boundary violations are the violations counted across the 737 failing seals. That record's 418 root violations led it to suggest that some sign readings have the wrong vowel class. Two known sources of false violations exist — slash-notation alternative readings (e.g., "ay/ā") and grammatical particles that follow words of any vowel class — but they have not been shown to account for the Phase-154 result. We therefore withdraw the earlier description of V12 as a settled methodology artifact: vowel harmony is an open problem for the reading set, and root harmony within single morphemes is in any case not testable with the current toolkit without morphological segmentation. No one of these three procedures is currently the program's authoritative V12 test: the records disagree because the test was redefined between runs, and the full rule changes are not recorded for every version. The V12 warning therefore stands as unresolved at every strength the three versions give (pass rates 75.3%, 64.2% and 36.1%), and no reading in this paper is either cleared or condemned by V12 until one protocol is frozen and re-run on the current 161-sign set.

3.18 Tamil-Brahmi Terminal Comparison (Phase-155)

H+M signs with strictly TERMINAL-dominant positional profiles (t_rate >= 0.50): 7 signs. All seven, with their terminal rates and the Phase-155 record's match decision, are: M233 = ūr (0.538, no match), M051 = pū/puḷ (0.515, no match), M065 = kuTam (0.545, matches the -am category), M305 = ōṭu/comitative (0.976, matches the -oṭu category), M125 = vil (0.561, matches the -il category), M249 = tii (0.959, no match), and M220 = al (0.984, matches the -al category). Testing these against the Tamil-Brahmi epigraphy terminal phoneme inventory (11 categories: -an/-aṉ, -ār/-ar, -ai, -iṉ/-in, -il, -ku/-kku, -um, -oṭu, -al/-āl, -am, -ām): 4/7 match (57%). No published source for that 11-category inventory is cited in this paper or in the Phase-155 record; the inventory is the test's own working list, and the match count can be checked against the record but not against an external authority. The non-matching signs (M233=ūr, M051=pū/puḷ, M249=tii) are content nouns that appear terminally in short inscriptions, not grammatical suffixes — their terminal position is structural (sole sign in a 2-sign formula) rather than morphological. A second, broader count was made after seeing this result, adding further suffix signs; earlier drafts reported it as reaching 8 of the 11 categories. Its derivation is not fully recorded — one sign named in the expansion (M220) was already in the original seven — so it is not reported as a result here. The 4/7 result above is the only recorded result of this comparison. The 11-category Tamil-Brahmi terminal inventory used in the test is the list printed in this section; no published source for that inventory is cited in this paper or in the Phase-155 record. Two cautions apply to the whole comparison: Tamil-Brahmi is roughly two thousand years later than the Indus material, so agreement can show only that the proposed endings are of kinds later attested in Dravidian epigraphy — it cannot validate the readings; and the V07 phonotactic violation rate under the stricter redefined test is 23.4% (270 of 1,154 decoded seals, Phase-155), concentrated in high-frequency signs. No record supports excusing those violations as mostly short abbreviated seals, and we do not do so here. The signs driving the V07 rate are recorded in the Phase-155 record, Part C: M328 (53 violations), M211 (49), M176 (46) and M342 (45) — three of them headline signs of

this model. This finding applies to the reading set as a whole: it bears on the sibilant notes in §3.11 and on §3.15's slot check, which use the same set, and on the Tier-2 sentence in §5.

3.19 Gulf Seal Fish-Sign Test (Phase-156)

This test extends §3.5 to the Gulf material. It tests the same published model of Avishai Roif (2025a, 2025b): the fish sign as a guild/title and maritime marker — in his expanded treatment, the "coastal controller" reading, with fish, boat, and jar signs recurring at port sites (Roif 2025b). On that model, fish signs in Gulf formal-seal contexts should likewise appear in compound, never as solitary markers. The Gulf counts and the verdict below are this program's own measurements.

Validation path §4.5, item 2, was executed using Laursen (2010) *Arabian Archaeology and Epigraphy* 21:96-134 and Mitchell (1986) in *Bahrain Through the Ages*. The Gulf extension searched about 145,000 characters of extracted text in the two source files named in the Phase-156 record: the extracted text of Laursen (2010) and the extracted text of the Bahrain volume (Mitchell 1986). The record names those files and the hit counts, but it does not record the search terms used, any term list, or an enumeration of the 27 Indus-script contexts, so the search cannot be re-run from the record as it stands. That text search found no mention of an isolated fish sign — and, to be precise about what a text search can show, no fish-sign mentions at all in the Laursen text (Phase-156: 0 hits). An absence in catalogue prose is weaker evidence than inspection of the seal images, and we report it as such: the Gulf material provides no counterexample to the compound-only pattern, but it does not independently confirm it. Parpola's (1994) Appendix lists compounds ending in *mīn* (4 attestations); an appendix of compounds cannot speak to isolated cases either way. The corpus result therefore stands on the Holdat count alone (0 isolated among 113 fish-bearing seals). The 27 Indus-script contexts in Laursen's catalog, in which likewise no isolated fish sign is reported, count a different unit (catalog contexts, not seals). They are not combined with the Holdat count, no combined figure is used anywhere in this paper, and the two results should always be cited separately: 0 isolated among 113 fish-bearing Holdat seals (§3.5), and no fish-sign mentions at all in the Laursen catalog text (this section).

3.20 Wells 2015 Sign List Cross-Reference (Phase-157)

Mining Wells & Fuls (2015) *The Archaeology and Epigraphy of Indus Writing* (161 pages): 284 sign references, **96 Dravidian language claims**, 403-reference positional analysis appendix (Appendix II). On this program's text mining of Wells, his position is compatible with a proto-Dravidian reading and with consistent sign positional classes; we cite this as our mining summary of his book, not as a quotation. (An earlier draft also claimed a Dholavira place-name convergence in two of Wells's sections; we could not locate that claim in our records and it is withdrawn.)

3.21 Mahadevan Terminal Ideograms + Grammar (Phase-158)

Mining Mahadevan (1982) *Terminal Ideograms in the Indus Script* and Mahadevan (1986) *Towards a Grammar of the Indus Texts*: the grammar paper contains 79 positional-class references and 61 passages matching our 3-slot model's slot terminology (classifier/title/suffix terms). These are terminology overlaps counted by this program's mining; they are not a count of passages shown, sign by sign, to agree with this model's slot definitions, and §3.23's per-paper counts use a different rule and must not be compared or added to them (see the reconciliation below). These Phase-158 counts and the Phase-160 counts in §3.23 are not the same measurement and must not be compared or added: Phase-158 counted every positional-class reference in the grammar paper (79) and every passage matching the model's slot terminology (61), while Phase-160 counted, per paper, only passages coded as grammar-support hits (8 for the same grammar paper). The two rules count overlapping material

differently. The mining recorded no vowel-harmony references in the grammar paper (Phase-158: `vowel_harmony_refs` = 0). That is an absence in one mined paper; it says nothing about the status of V12, which remains unresolved (§3.17).

3.22 Parpola 1994 Reading Overlap Count (Phase-159)

Mining Parpola (1994) *Deciphering the Indus Script* (1,566,507 chars): **44/75 HIGH-confidence readings appear in Parpola's text** (59%). This is not independent cross-validation: Parpola's framework guided the choice of readings (§7), so agreement with his book is partly expected by construction, and the count records appearances in his text, not a sign-by-sign check that he proposed the same reading for the same sign. We report it as a measure of how much of the HIGH set sits within the Parpola tradition. Two further counts from the same mining are withdrawn as evidence: the 6,869 'genitive/possessive' text hits (the recorded sample contexts are front matter and other noise, and the hits were never coded for relevance to M267) and the 47 'classifier/determinative' references (not located in the Phase-159 record, and generic mentions would not support the specific INITIAL-slot model in any case).

3.23 Mahadevan Four-Decade Grammar Consistency (Phase-160)

Mining the Mahadevan papers in this program's text corpus (the aggregate counts '10 papers, 1972–2018' and, in §3.24, '38 papers' are this program's own corpus counts and could not be tied to a verifiable external source): the papers actually mined record positional-grammar hits compatible with the 3-slot model — Grammar of Indus Texts (1986): 8 hits; What Do We Know (1989): 7 hits; Terminal Ideograms (1982): 6 hits. These are one scholar's works across four decades, not independent witnesses, and 'consistent with' here means only that the mining found classifier/title/suffix terminology and positional classes of the kinds the model uses. The disclosure in this section — that these are this program's own text-mining counts, internal outputs, not externally sourced figures — applies equally to §§3.20–3.22 and §3.24. The akam-puram (2011) paper identifies a crescent-moon sign as an 'outer city' marker at INITIAL position; this program's mining records that as compatible in position with our INITIAL class, but no sign-number mapping between that paper's sign and any sign in our inventory is printed or recorded, so no stronger consistency is claimed. Place Signs (1981) documents 10 settlement/ūr sign references in the mining count; that count does not by itself check this paper's specific M233=ūr assignment, and it is reported as a mining count only. Across these works — one scholar's papers over four decades, not independent witnesses — the mining found the same kinds of positional terminology throughout. For the record, the Phase-160 counts give per-paper hits for all ten papers mined: Grammar of Indus Texts (1986) 8; What Do We Know (1989) 7; Terminal Ideograms (1982) 6; Encyclopaedia (1994) 5; Place Signs (1981) 4; Akam-Puram (2011) 4; Toponyms (2018) 3; Bilingual Parallels (1972) 3; Murukan (1999) 2; Blue Neck (2008) 2.

3.24 Literature Extraction Ceiling (Phase-161/162/165)

Systematic mining of Parpola (1994), 38 Mahadevan papers (1970–2018), and Wells & Fuls (2015) for sign-reading proposals yielded **0 new direct M-number reading assignments** against the LOW-confidence sign set (240 signs at time of Phase-161 analysis; 236 in the current anchor table after 4 sibilant promotions). No new direct M-number reading assignments for the LOW set were found by this pipeline. That is a null result for this search method only: its recall has never been measured against a hand-checked list of known proposals, and the same mining did surface proposals for signs outside the LOW set, so the null should not be read as proof that the H+M set is at the frontier of published scholarship. It is consistent with that view, at the strength a single unbenchmarked search can support. The large majority of LOW signs carry placeholder readings for which this mining run found no proposed phonetic value in the literature it searched. Because the run's

recall was never measured, that statement covers the searched literature as this pipeline read it — not all available reference literature. The 18 signs in Appendix B resisting MEDIUM promotion (after M198 was promoted to PROVISIONAL_MEDIUM in Phase-166) may be personal-name syllable candidates; the statistics alone do not distinguish them from any other medial-only sign class (§3.9), and current methods did not resolve them. A bilingual text, or a corpus large enough to identify repeated name sequences, would be needed. The text-mining and positional pipeline used here did not advance the LOW-sign set further on this corpus. That is a statement about this pipeline, not about the material: rival-model comparison, expert coding, and image inspection remain open on the same corpus (§3.30), and the ICIT corpus (5,318 inscriptions vs 1,670) or a new archaeological find would add evidence of kinds this pipeline cannot supply.

3.25 Sibilant Exploratory Promotions (Phase-163)

Mining all three reference sources specifically for sibilant-initial readings (ca-/ce-/ci-/co-/cu-) near sign references: 15 /cu/co/ca/ candidates identified for LOW signs via text proximity analysis. Four signs promoted to PROVISIONAL_MEDIUM status based on >=3 text-proximity hits in the mined literature (a hit means the proposed reading appears near the sign's number in OCR'd text; it is not a verified assignment of that reading to that sign): M165 ('cul', 4 Parpola references), M330 ('can', 4 Parpola references), M202 ('can', 4 Mahadevan references), M198 ('co', 3 Parpola references; upgraded to PROVISIONAL_MEDIUM in Phase-166). These are **exploratory** — text proximity in OCR'd academic text does not guarantee a direct sign assignment; they require expert review. With these 4 additions: H+M = 161 (was 157), token coverage = 90.96% (was 90.75%), H+M-candidate seal coverage = 69.8% (was 69.1%).

3.26 Meluhhan Name Phonological Matching (Phase-164)

Testing 6 attested Meluhhan personal names from Mesopotamian cuneiform against corpus sign sequences: no strong matches (>=3/4 phonological slots) found in the 1,670-seal corpus. Partial matches (2/3 slots): Urgula (*ur-gu-la*) — 17 seals with M233(ūr) + ku-class + suffix; Nanna-a (*nan-na-a*) — 9 seals with nal/naN + genitive + suffix. These are below the confidence threshold for a personal name decipherment claim. The 1,670-seal corpus is insufficient — ICIT (5,318 texts) would provide ~3x more evidence for low-frequency MEDIAL sign sequences required for personal name identification. Per-name results are in the Phase-164 record (`backend/reports/phase164_meluhhan_names.json`).

3.27 Sibilant DEDR Check (Phase-166)

A within-model plausibility check of the four Phase-163 provisional MEDIUM sibilant upgrades: whether each proposed reading has a DEDR entry, a positional profile consistent with its class, and a form allowed by Proto-Dravidian phonotactics. This check uses the same framework that proposed the readings; it is not cross-validation and not independent confirmation. **Results**: M330=*can* PROVISIONAL_MEDIUM (DEDR 2322, **can-* 'to do; matter, affair'; INITIAL position; 3 corpus tokens; 4 Parpola references); M165=*cul* PROVISIONAL_MEDIUM (DEDR 2700, **cul-* 'whirl, depth'; MEDIAL position; 2 corpus tokens; 4 Parpola references); M202=*can* PROVISIONAL_MEDIUM (DEDR 2322; MEDIAL position; 4 corpus tokens; 4 Mahadevan references); M198=*co* PROVISIONAL_MEDIUM (DEDR 2816, **co-* 'to say, speech'; INITIAL position; 6 corpus tokens; 3 references — it met the >=3 promotion threshold of Phase-163 but not the >=4 threshold Phase-166 uses for a 'strong' reference count; all four signs remain PROVISIONAL_MEDIUM on the same basis, and the difference in reference counts changes no sign's category). All four upgrades have DEDR backing and Proto-Dravidian phonotactic validity (*c*-initial is a valid Proto-Dravidian consonant onset; Krishnamurti 2003). The outcome is that all four assignments remain PROVISIONAL_MEDIUM, unchanged: each reading is plausible

within the model and none is verified as a published assignment of that reading to that sign. Expert review is required before any promotion. (The Phase-166 record's label for this outcome, PROVISIONAL_SUPPORTED_WITHIN_MODEL, is not used in this paper, for the same reason as in §3.15: every underlying assignment is unverified, and a label containing 'supported' invites misreading.)

3.28 Expanded Meluhhan Name Matching (Phase-167)

Expanding Phase-164 from 6 to 25 attested Ur III Meluhhan personal names (Parpola 1975; Steinkeller 1982; Potts 1994; Reade 2001; Kjaerum 1983) and testing phonological segment matching against all 1,670 Holdat seals using the 161-anchor reading set. The matching algorithm was intentionally broad. In the Phase-167 record (`backend/reports/phase167_meluhhan_names_expanded.json`), each name is pre-split into syllable-like segments (for example, Apil-Adad as a / pil / a / tat); a seal counts as matching a name when at least 50% of the name's segments are matched by the seal's decoded readings — counted segment by segment, not as whole names — and the record's 'strong' band means at least 75%. The record does not state a fuller rule for how a single segment counts as matched, and none is claimed here. All 25 names produced apparent 'matches' at this threshold due to the prevalence of common short readings (*ā*, *kol*, *il*) in the H+M-covered seal set. **Effective result: NO_DISCRIMINATIVE_MATCH** — consistent with Phase-164. Personal name decipherment using the current 1,670-seal corpus is not feasible. The ICIT corpus (5,318 texts) with contextual metadata is required to identify name-bearing seal sequences.

3.29 Decode-Blocker Statistical Analysis (Phase-168)

Statistical assessment of the top 19 decode-blocking LOW signs from Phase-130 (signs whose presence in a seal prevents full H+M coverage). Each blocker's existing LOW reading was assessed for: (1) phonotactic validity in Proto-Dravidian; (2) positional profile consistency; (3) Phase-110 SA evidence where available.

The 19 signs, with their corpus frequencies from the Phase-168 record, are: M270 (6 tokens) and M386 (6); and, at 5 tokens each, M183, M365, M412, M151, M304, M321, M357, M137, M239, M223, M254, M143, M345, M329, M190, M295 and M402 — 97 tokens in total. The 19 are the 18 Appendix-B signs plus M412. The coverage figure below is therefore checkable from this page: 6,369 + 97 = 6,466 tokens, and 6,466 / 7,002 = 92.35%. **Results**: 19/19 blocker LOW readings are phonotactically plausible (all use valid Proto-Dravidian initial consonants). If these 19 blockers were promoted to MEDIUM, token coverage would rise from 90.96% to an estimated 92.35% — a hypothetical figure, reported here only to size the blockers' effect. Note: the planned full-corpus SA run with 161 pinned anchors could not be executed (a constraint-initialisation incompatibility on the available platform), and the statistical assessment used instead does not substitute for it: it checks plausibility (valid Dravidian initial consonants, positional consistency) and nothing more. In light of the Phase-107 result on the SA machinery (§2.4), no SA-based verdict is claimed for these signs. **These 19 signs remain at LOW confidence** — phonotactic plausibility is a necessary but not sufficient condition for MEDIUM promotion. Individual DEDR cross-validation (as in Phase-166) would be required for each.

3.30 Master Evidence Synthesis (Phase-169)

Updated Phase-141 evidence scorecard incorporating Phases 142–168. **Results**: 32 total evidence items (up from 23 at Phase-141), in the Phase-169 record's own categories: 12 STRUCTURAL, 8 LINGUISTIC, 8 EXTERNAL, 4 DECIPHERMENT. These ratings are assigned by this program, on labels defined by this program; the earlier draft's '79.8% aggregate confidence score' is withdrawn, because no weighting rule for it was ever recorded, so the number carried no meaning. The items mix two

different kinds of check — internal consistency checks (which can show the model agrees with itself) and external comparisons (which can test it against something outside) — and only the second kind is evidence about the world. The full per-item list, with each item's label, is in the Phase-169 record (`backend/reports/phase169_master_synthesis.json`); it is not printed in this paper, and no single tally at the top labels is printed, because any such tally would mix the two kinds. Among the new items: Gulf seal text search (E06 — no fish-sign mentions found; no counterexample, not corroboration, §3.19), Wells 2015 mining summary (E07 — this program's text-mining summary of Wells's book, not a quotation and not a sign-by-sign check; the mining's coding rules and recall were not validated, §3.23), Mahadevan grammar text-mining comparison (L07), Parpola 1994 reading overlap count, 44/75 = 59% (E08), literature extraction null (S09 — a null for this mining pipeline only, recall unmeasured, §3.24), sibilant DEDR check (L08), adversarial challenge battery (S10 — an internal battery run by this program against its own claims; the 10 challenges, their pass criteria, and the caveat attached to each of the 6 caveated survivals are in the Phase-149 record and are not summarised here): of 10 challenges, 4 survived outright and 6 survived with caveats, none failed outright (Phase-149). S10 is an internal consistency item, not an external comparison, and is counted with the internal items wherever §3.30's counts are used, M267 chi-square motif-independence (S11), and blocker phonotactic plausibility (S12 — a plausibility check, listed for completeness; it is not evidence that the blocker readings are correct, §3.29). (Category counts as in the Results paragraph above: 12 STRUCTURAL, 8 LINGUISTIC, 8 EXTERNAL, 4 DECIPHERMENT — printed once, there.) No further advance was found from the text-mining and positional methods of Phases 142–168 on the current 1,670-seal Holdat corpus and the published literature mined for them. That is a statement about those methods, not about the corpus: other approaches — rival-model comparison, expert coding, image inspection — remain open on the same material, and this program's own later re-audit of its anchors (Phases 107–109) was itself an advance made on this corpus.

3.31 Grammar Variance Retest at 161 H+M (Phase-170)

Retest of the Phase-133 grammar explained variance metric with the updated 161 H+M anchor set (adds M330, M165, M202, M198). Method: for each H+M sign with >=3 corpus tokens (147/161 qualify), compute I/M/T positional rates and test whether the 3-slot grammar model (TERMINAL if T>=0.60, INITIAL if I>=0.50, MEDIAL if M>=0.65) correctly classifies the sign's dominant positional class. The Phase-170 record states the rules in the order TERMINAL, INITIAL, MEDIAL, else MIXED; it does not state what happens when a sign's rates satisfy more than one threshold, and that tie/overlap handling is not on record. **Results**: 137/147 signs (93.2%) are classified by the thresholds into the positional class that is already their dominant class. This is a self-consistency check, not an accuracy measurement: the thresholds (TERMINAL if T>=0.60, INITIAL if I>=0.50, MEDIAL if M>=0.65 — program conventions, with no sensitivity analysis on record) are applied to the same rates that define each sign's dominant class, so a high figure is expected by construction and the check can only catch internal contradictions. It is not comparable to Phase-133's figures, which measured how well the grammar pattern matched whole inscriptions: full match 5.1% of seals and partial match 38.4%, together 43.5%. Phase-133's 44.3% is a different quantity in the same record — its sign-position accuracy (explained variance) — not the sum of the full and partial rates. The two numbers measure different things, and neither confirms the other. The evidential weight for the grammar model rests on the Phase-134 permutation and held-out tests, not on this retest.

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4. Discussion

4.1 Proposed Readings Under the Candidate Model

The three sequences below are illustrative sequences, built from the model's readings to show how the grammar is meant to be read. They are sign-number sequences; this paper does not tie them to identified, catalogued seals, and they should not be cited as transcriptions of specific seals.

Under the grammar model, a typical seal reads as:

- **M211 M099 M342** = *kol-kol-ay* = "the jar-guild lord" (unicorn seal, formal guild title)
- **M062 M342 M176** = *erutu-ay-an* = "the bull's man" (zebu bull seal, personal name)
- **M045 M176 M267 M328** = *yānai-an-iN-āl* = "the elephant guild man's man" (compound title)

Under the model — and only under the model — these inscriptions read as a directory of guild members rather than a commodity ledger; the two readings have not been separated by any test yet (§4.5 names a future one), and rival functions (ownership, clan, cultic, or commodity marking) are not ruled out. The translations above carry the uncertainty of every reading they use; the first, for example, depends on two different signs (M211 and M099) both being read kol. On repeated readings, the rule of this model is: homophony is permitted — two different signs may carry the same reading — whenever each sign's own evidence supports that reading, as with M211 and M099 here. Shared readings are common in the 161-set, but no audited count of them is on record, and none is claimed. The Arthaśāstra (a layered Sanskrit text whose core is usually placed in the last centuries BCE, roughly a millennium and a half after the end of the IVC period) documents Sanskrit superintendent titles of a broadly similar kind — *koṣṭhāgārādhyakṣa* (storehouse superintendent), *nāvadhyakṣa* (ship superintendent). No word is shared with the proposed Dravidian readings, and the parallel is an analogy about administrative practice, not evidence of continuity, which remains speculative.

4.2 The Substrate Evidence

Two signs (M374=kul, M351=vī, six tokens each) are assigned candidate Munda substrate readings under the model (see §3.3 for the limits of both). We record, as speculation and nothing more, the scenario they would fit if they were correct: a linguistically mixed IVC in which a Dravidian-writing administration absorbed loanwords from Munda-speaking communities. Two candidate signs cannot establish that scenario, and no claim about an administrative elite or about community structure is made on their basis. The Munda substrate in historical Dravidian and Indo-Aryan languages is well-attested (Witzel 1999; Southworth 2005). If the proposed readings are correct, this would suggest that such substrate influence predates later attested Munda vocabulary.

4.3 The 18 Signs Resisting MEDIUM Promotion

Eighteen signs with corpus frequency 5–6 resist MEDIUM promotion under current methodology. Their modals from the syllabic LM are syllables without clear DEDR matches (e.g., 'o', 'e', 'pi', 'du', 'pit'). One reading of this pattern, which the model adopts, is that these signs are syllabic components of personal names — in a system where personal-name syllables could be arbitrary, they would not be recoverable from distributional statistics alone. The statistics do not distinguish that reading from any other medial-only sign class (§3.9 above), so it is a preference of the model, not a finding. The limit stated here belongs to these distributional statistics, not to method in general: other evidence (repeated name sequences, external name matches) could separate the two, and none is available in this corpus.

4.4 Limitations

1. **No bilingual text**: All readings remain provisional until a bilingual inscription (IVS + Sumerian, Akkadian, or Sanskrit) is discovered. No numerical probability attaches to any individual reading; the confidence labels are rule-based categories (§2.2).
2. **Corpus size**: 7,002 tokens is small for linguistic reconstruction. High-frequency signs have more positional data than low-frequency signs, but frequency has not made their readings secure: the V07 phonotactic violations are concentrated in high-frequency headline signs (§3.18), and the program's largest single correction was to a 400-token sign (§3.4). Low-frequency signs (the irresolvable 18) have too little data for either kind of check.
3. **Single corpus**: The Holdat corpus covers formal stamp seals only. Copper tablets, potsherd graffiti, and the Dholavira signboard require separate analysis.
4. **SA simulation**: The syllabic language model is described in this program's records as trained on modern Tamil material, but no record names the specific corpus, the syllable-inventory source, or the size of the training data, so the training set cannot be reconstructed (§2.4). Phonological drift over 4,000 years may introduce systematic errors, and the size of that risk cannot be assessed without the training specification.
5. **Confidence standards**: The MEDIUM threshold is a rule, not a measured standard: it has no known false-positive rate (§2.2), and it admits routes — SA+DEDR agreement and, for four signs, literature text proximity — whose error rates are likewise unmeasured. Some assigned LOW readings may be correct but lack sufficient support under the rule.
6. **Language-family baseline**: The present model tests Proto-Dravidian compatibility more extensively than competing baselines. A formal comparison against Proto-Munda, early Indo-Aryan, and language-neutral syllabic models remains necessary before the linguistic interpretation can be treated as discriminative rather than merely compatible with Proto-Dravidian. For the record, the comparisons that do exist are mixed: a syllabic language-model comparison favoured Dravidian over Sanskrit on the 157-sign set (Phase-134 F12), while a phonological-profile comparison of the HIGH anchors found Dravidian and Sanskrit fits equal and rated the question AMBIGUOUS (Phase-134 F3).

4.5 Validation Path

The candidate model is falsifiable in principle at the first of the following points. None of the three is a test that can be run on the data available today: the first requires a future discovery, the second is complete and did not confirm the model, and the third has no discriminating prediction on record yet. The three points are stated so that a reader can see exactly what would have to happen, and what has already happened, at each. 1. **Discovery of a bilingual text**: Any Indus inscription alongside a known script (Sumerian, Akkadian, Elamite) at a Gulf trade site would allow direct validation of individual sign readings. 2. **Gulf material** (Phase-156, now complete): the Laursen (2010) Gulf Type seal catalog text and the Mitchell (1986) Bahrain volume text were searched. The search found no fish-sign mentions at all, so the Gulf material provides no counterexample to the compound-only pattern — and, because the search found no fish signs in the text, no confirmation of it either (§3.19). The compound-only verdict rests on the Holdat corpus count (0 isolated among 113 fish-bearing seals); no extended verdict across both corpora is claimed. 3. **Radiocarbon-dated stratigraphic contexts**: a future test of this kind could separate the guild-identity interpretation from the commodity-tally model only if the two are first made to predict different things — for example, different sign-sequence patterns in seals from a single dated layer. Consistent 'guild' readings in one layer, on their own, would presuppose the readings and would support neither model over the other. No such discriminating prediction is yet on record, so this point is a direction for future work, not a falsification test of the present model.

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5. Conclusion

We propose a reproducible computational grammar and candidate Proto-Dravidian anchor model for the Indus Valley Script. The strongest findings, which can be stated with high confidence, are two: robust non-random positional structure ($z = 10.3$; no permutation in 2,000 exceeded the observed statistic ($p < 0.001$; 0 of the 2,000 permutations exceeded the observed statistic, so the plain estimate is 0/2,000 and the standard add-one estimate is (0+1)/(2,000+1) = 0.0005, the smallest non-zero value a test of this size can report) — §2.3 defines the statistic), and the correction of the M267 misassignment (M267 is not the fish sign: it occurs across all motif types, §3.4). Two further structural results are stated at the strength their tests support: the fish-sign compound-only pattern in the Holdat corpus (0 isolated among 113 fish-bearing seals; the Gulf extension is weaker, §3.19), and a recurring three-slot positional pattern whose positional classes are measured but whose semantic labels (classifier, title, name) are this model's interpretation (§3.2). The candidate Proto-Dravidian readings (161 H+M signs; 90.96% token coverage — assignment coverage in the sense of §2.2: the share of tokens whose sign carries a candidate reading, not a share of verified readings) are Tier 2 claims supported by multiple tests. Those tests are not independent of each other — most are computed from the same corpus, and several from the same seals (§2.2) — the HIGH/MEDIUM labels are rule-based categories with no measured error rate, and the set also carries the unresolved V12 warning and the 23.4% V07 phonotactic violation rate (§§3.17–3.18) — a rate concentrated in high-frequency headline signs (M328, M211, M176 and M342 drive it, §3.18), not spread over rare or doubtful readings. The readings have not yet been externally reviewed. Speculative interpretations — guild-identity semantics, the two DEDR-backed readings for which a Munda derivation was proposed but not established, individual seal translations — are Tier 3 claims requiring expert review and explicit caveats.

This study does not claim epigraphic finality in the absence of a bilingual inscription. The corpus shows non-random positional structure of the kind a writing system produces and also of the kind some non-linguistic marking systems produce; the tests in this paper do not separate those two, and the frequency-curve evidence that might have spoken to the question was uninformative (§2.1.1). Within that limit, the evidence favours a Proto-Dravidian reading framework under the tested assumptions. The ICIT corpus is the immediate priority for testing whether the candidate model generalises to a larger public corpus. All phase reports, scripts, and the anchor table are available in the Glossa-Lab repository.

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6. Data Availability

All sign readings, confidence levels, DEDR citations, basis statements, analysis scripts, and phase reports are available in the Glossa-Lab repository (https://github.com/BitConcepts/glossa-lab; the version of the paper you are reading is on branch `preprint-v8-f2-spec068`, commit `196ea0a2`). The DOI named on the title page identifies this preprint and its version record on Zenodo. The exact text of this draft is identified by the branch and commit named here, not by the DOI. The supplemental files named in this paper (the anchor table, the status audit, and the CSV supplements in `research/indus/supplemental/`) are in the same repository at the same commit; no separate archive copy of them exists yet. The commit named here is the revision commit: the commit at which this paper's text was frozen for the current round. Commits after it on this branch add only records — the rebuilt PDF, the version note, and this pin itself — and do not change the paper's text. A reader who checks out the named commit therefore sees the paper's text exactly as frozen; the only sentence that differs in that checkout is this one, which there names the previous round's commit. The anchor table — all 161 HIGH/MEDIUM rows, with each sign's reading, DEDR citation, and basis — is `research/indus/anchor_table.csv`; the full 397-sign inventory, with each sign's status and corpus frequency, is `glossa-corpus/indus/ANCHOR_STATUS_AUDIT.csv`; per-phase results are in `backend/`

reports/. Neither table is printed in this paper; both are in the repository, on the branch and at the commit named above. Both tables are the paper's central evidence and are deliberately not summarised by selected rows anywhere in this paper: a reader who wants to check any single reading should go to the anchor table itself, at the commit named in §6, rather than to the examples in §3.3, which are illustrative only. The Holdat LLC Indus Corpus v3 is not publicly available at time of writing. Because the Holdat corpus is not yet publicly released, full reproduction of corpus-level counts requires either access to that corpus under agreement or re-running the pipeline on a compatible public corpus after sign-code crosswalking. Supplemental datasets (fish-sign compound-context listing, iconographic formula enrichment table, formula bigram table, and positional grammar permutation summary) are available in research/indus/supplemental/.

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7. Acknowledgments

Mahadevan (1977) for the sign catalogue. Parpola (1994, 2010) for the Dravidian decipherment framework that grounds all readings. I thank G. Martini for discussion of later South Asian administrative vocabulary. Ashish Nair for the independent scorecard study (arXiv:2604.17828). Avishai Roif, as the source of the fish-sign model the tests in §3.5 and §3.19 were built on: the fish sign read /mi/ (mīn) as a value/divine marker and guild/title marker, and the "coastal controller" maritime-guild reading (Roif 2025a, 2025b). The model and its predictions are his; the measurements in those sections are this program's own.

AI Disclosure: This research was conducted with AI-assisted computational tooling (Glossa-Lab pipeline, Warp/Oz agent). Analysis scripts, anchor tables, and phase reports are available in the Glossa-Lab repository. The Holdat LLC Indus Corpus v3 is not publicly available; full corpus-level replication requires access to that corpus or a compatible public corpus such as ICIT after sign-code crosswalking. Statistical tests were designed, executed, and interpreted by the author; AI tooling was used for scripting, data management, and literature search — including the text-mining counts reported in §§3.21–3.24 and the text-proximity hits behind the four PROVISIONAL_MEDIUM promotions (§2.2). Those machine-produced counts were checked by the author against the phase records at the level of totals and sampled contexts, not by a full hand recount, and no benchmark of the mining pipeline's recall or precision exists (§3.24). Model names, versions, prompts, and run settings for the mining steps were not archived with the phase records; the counts are therefore reproducible from the records' outputs, not from a re-runnable specification. No AI tool assigned a confidence label, a DEDR candidate, or a verdict in this program: those were fixed by the rules and records cited at each step.

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Appendix A: Summary of New Phase-128/129 Anchors

Sign	Reading	Confidence	DEDR	Evidence
M374	kul	MEDIUM	1709	DEDR 1709 (kulam, clan/lineage); a Munda *kul* proposed by this program (Phase-128) has no Munda source cited in our records
M351	vī	MEDIUM	5388	DEDR 5388 (vī, seed); the asserted derivation from Munda *bi* by a 'bilabial shift' is unestablished
M072	mā	MEDIUM	4751	INITIAL-only (12/12 tokens); collocates peN, veL, comitative
M149	or	MEDIUM	987	MEDIAL; [kōn̲] -> or -> [an]; Dravidian oru=one/a certain
M185	pul	MEDIUM	4336	MEDIAL; descriptor before āl/man; pul=humble/grass

Appendix B: The 18 Irresolvable Signs

These 18 signs have corpus frequencies of 5 (sixteen signs) or 6 (M270, M386) — 92 tokens in total, 1.31% of corpus tokens — appear in MEDIAL position in the Phase-144 blocker analysis (personal-name slots in the model), have SA modals without DEDR matches, and resist substrate identification.

Each of them blocks the 5 or 6 seals it occurs on, so together they account for at most 92 of the 505 seals not fully covered by H+M candidate readings (§3.6); the rest are blocked by other LOW signs.

Signs: M183, M190, M223, M239, M254, M270, M295, M304, M321, M329, M345, M357, M365, M386, M137, M143, M151, M402. (M198 removed — promoted to PROVISIONAL_MEDIUM by Phase-166.)

Working hypothesis: These are syllabic components of personal names — arbitrary sound sequences used to encode individual names within the guild-identity system. Their decipherment requires either a bilingual text or a corpus large enough to identify repeated name sequences across related inscriptions.

Appendix C: Foundation Validation Summary

Internal consistency suite (foundation_check.py): on the October 2026 execution tree, the changed suite reported 40 passed, 0 failed, 8 warnings. The earlier May 2026 result — 59 checks, all passing, at Phase-170 — belongs to the previous version of the suite and is recorded here as provenance, not as the current result. These are internal consistency checks, not external validation. Items from the May 2026 suite include: - Dravidian Tier 1 lift ratio: 3.13x above null (Phase-44) - Contact zone HIGH anchors: supported within internal checks (Phase-46) - Sign coverage: 397 M-numbers all addressed - Dashboard coverage metrics: guaranteed <=1.0 (no inflation) - Grammar model: the three positional classes are internally consistent under the suite's checks; the classifier/title/suffix meanings are the model's interpretation and are not checked by the suite - Fish sign isolation: 0 isolated among 113 fish-bearing corpus seals (Phases 124/127); the Gulf text search (Phase-156) found no fish-sign mentions at all and contributes no denominator (see §3.19) - Sign accounting: 390 attested corpus signs; 397 Mahadevan catalogue signs; 7 catalogue-only signs with zero corpus tokens

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End of Preprint v8
Glossa-Lab, May 2026; revised October 2026
Word count: 17,016